CHEN Muyang

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Multimodal large language models · Multimodal reasoning · Affective computing

Education

The Chinese University of Hong Kong, Shenzhen Sep 2023 – Jul 2027 (expected)
B.Eng. in Electrical and Computer Engineering (Computer Engineering Stream)

Cumulative GPA: **3.81/4.00** · Major GPA: **3.90/4.00** (**3/60, Top 5%**, Computer Engineering stream)
Selected coursework (all A): Machine Learning; Machine Intelligence and Applications; Linear Algebra; Parallel Programming; Image Processing and Computer Vision; Probability and Statistics I

Manuscripts

MISID: A Multimodal Multi-turn Dataset for Complex Intent Recognition in Strategic Deception Games.
Shufang Lin*, **Muyang Chen***, Xiabing Zhou, Rongrong Zhang, Dayou Zhang, and Fangxin Wang.
Under review at AAAI 2027 · *Equal contribution

Research Experience

MISID – Multimodal Hidden-Intent Recognition Sep 2025 – Present
Undergraduate Researcher, NAIS Lab, CUHK-Shenzhen | **Co-first author**
Lab PI: Prof. Fangxin Wang | Research mentor: Dr. Dayou Zhang

- Co-designed a two-tier annotation scheme for 3,962 aligned audiovisual utterance segments from multi-turn strategic-deception games, linking utterance-level role and affect labels with cross-turn factual evidence and cross-modal inconsistencies. Independently designed the inter-annotator agreement (IAA) evaluation protocol and conducted the assessment (Krippendorff's α: 0.80–0.94 across structured fields).
- Contributed to the zero-shot evaluation of 19 language and video-language models, conducted ablation studies, and co-designed the LLM-as-a-Judge protocol.
- Co-designed FRACTAM, a reasoning framework integrating modality decoupling, long-range retrieval, and cross-modal evidence chains. The framework improved mean Hidden Intent Inference Score (HIS; 0–100 scale) from **34.22 to 42.79** across ten language-model backbones.
- Independently designed the **modal-decoupling and text-reconstruction component** for annotation and inference, using acoustic and facial-action features to generate separate audio and visual descriptions without speech transcripts to preserve potential cross-modal inconsistencies. In team ablations, adding this component alone raised HIS from **43.35 to 49.76** for Claude Sonnet 4.5 and from **36.35 to 41.80** for DeepSeek-R1.
- Revised inference prompts to guide models in identifying subtle contextual cues that distinguish alternative explanations for participants' behaviour, improving performance in multiple tests. On the full MISID benchmark, role-identification accuracy increased by **4.58 percentage points** on average across ten LLMs.

Visual Token Compression for Vision-Language Models Sep 2026 – Present
Final-Year Project, CUHK-Shenzhen | Supervisor: Prof. Fangxin Wang

- Investigating client-side visual token compression to reduce the bandwidth needed to transmit visual information to server-side vision-language models while preserving task-relevant information.
- Adapted a question-independent, MetaCompress-based compression module after the Qwen3.5-4B visual merger for image question answering; froze the backbone and trained the compressor on 20,000 images from GQA and VQA v2. Compared trained and untrained variants with random and uniform token selection.
- Evaluated visual-token selection and compression for video question answering with Qwen3.5-27B in preliminary tests on 40 StreamingBench questions; random selection matched full-input accuracy while retaining only **20% of the visual tokens**.

Research Experience (continued)

AffectiveArt – Artwork Emotion Understanding May – Jun 2026
Team Lead | ACM Multimedia 2026 Grand Challenge, Track 2

- Led a three-member team and designed the overall system architecture.
- Fine-tuned Gemma, Qwen, SigLIP2, and DINOv3 classifiers on the official EmoArt training split; independently designed and implemented **task-specific weighted voting** using softmax-normalised validation accuracies. The ensemble improved test accuracy over the best individual model for each task by **8.20, 5.12, and 4.33 percentage points** for emotion, valence, and arousal, respectively.
- Independently fine-tuned SigLIP2 and DINOv3 to predict the salience of visual attributes for affective judgements; the predicted affect labels and attribute salience conditioned a LoRA-tuned Qwen3.5-27B model to generate artwork emotion descriptions.
- The system placed **5th among 14 teams** in the challenge track, achieving 79.2% emotion, 88.3% valence, and 94.0% arousal accuracy on the hidden test set.

Selected Academic Projects

Interpretable Butterfly Classification Code Summer 2025
Project Lead | ECE4513 Image Processing and Computer Vision, CUHK-Shenzhen

- Led a three-member team to develop a CNN with four convolutional layers for ten visually similar butterfly species; achieved **83.23% test accuracy**.
- Implemented Class Activation Mapping (CAM) from scratch to visualise class-discriminative regions; applied PCA to final-layer feature maps to visualise and compare activation patterns across samples of the same species. Documented the analysis in a technical report.

Teaching Experience

Undergraduate Student Teaching Fellow Sep – Dec 2024
MAT1001 Calculus I, CUHK-Shenzhen

Supported weekly tutorials and office hours for approximately 60 students; collected student questions, explained selected homework problems, and invigilated quizzes.

Honors & Awards

Undergraduate Research Award (URA), CUHK-Shenzhen Jul – Sep 2026
Dean's List, School of Science and Engineering, CUHK-Shenzhen 2023–24, 2024–25, 2025–26
Academic Specialty Entrance Scholarship (Second Class) 2023–24 through 2026–27

Skills & Languages

Programming: Python, C++ | **Frameworks & Libraries:** PyTorch, OpenCV
Tools: Git, Linux, LaTeX
Languages: Mandarin (native); English (IELTS Academic 7.0 overall)